

Selection rules, entropic order parameters and the massive sector of the $N = 2$ flow $k \rightarrow k - 2$

Working note

Scope. Bosonic $N = 2$ coset (l, m, s) with the conventions of `N2minimal.m` (S-matrix eq. (2.6) of N2Gab normalised by $1/K$, $K \equiv k + 2$). I take k odd throughout (no subtleties with the invertible line $[k, 0, 0]$). For a node T I compute the commuting lines F_T and $\mu_T = (\sum_{F_T} d^2)^2$ (paul.pdf eq. (5)); $\lambda_T \equiv \sqrt{\mu_T} = \dim F_T$ is the index of T in the completion and the constant in the two-interval certainty relation $S_{\text{int}} + S_{\text{tw}} = \log \lambda_T$. The numbers come from `n2_nodes.py`, which I checked for $k = 3, \dots, 11$.

1 UV node of $(0, -2, 0)$ and IR node of its image

UV. The smallest node containing $\phi_{UV} = (0, -2, 0) \sim (k, k, 2)$ (and its conjugate) is

$$N_{UV} = \langle (0, -2, 0) \rangle = \{(0, 2n, 0)\}_{n \in \mathbb{Z}_K} \simeq \mathbb{Z}_K, \quad F_{UV} = \{(L, 0, S)\}_{L+S \text{ even}} \simeq \mathfrak{so}(3)_k \boxtimes \mathbb{Z}_4, \quad (1)$$

with $\mathbb{Z}_4 = \{(0, 0, 0), (0, K, \pm 1), (0, 0, 2)\}$ generated by $g = (0, K, 1)$, $g^2 = (0, 0, 2)$. Hence

$$\mu_{UV} = 16 (\dim \mathfrak{so}(3)_k)^2, \quad \dim \mathfrak{so}(3)_k = \sum_{L \text{ even}} d_L^2 = \frac{K}{4 \sin^2(\pi/K)}. \quad (2)$$

IR. The IR is approached by the D-term of the non-BPS field $(4, \pm 2, 0)$ of the $k' = k - 2$ model (N2Gab §5.3), with $\Delta = 2 + 10/k$. This field exists for $k' \geq 4$. Its node is

$$N_{IR} = \langle (4, 2, 0) \rangle_{k-2} = E_{k-2}, \quad E_k \equiv \{(l, m, 0) : l, m \text{ even}\}, \quad F_{IR} = \mathbb{Z}_4, \quad \mu_{IR} = 16. \quad (3)$$

The only $k - 2$ nodes that contain $(4, 2, 0)$ have $\mu \in \{1, 4, 16\}$.

k	$\dim \mathfrak{so}(3)_k$	$\mu(N_{UV})$	$\mu(E_k)$	$\mu(N_{IR})$	h_g (UV)	$h_{g'}$ (IR)
3	3.618	209.44	16	16*	7/8	3/8
5	9.296	1382.6	16	16*	3/8	7/8
7	19.23	5919.4	16	16	7/8	3/8
9	34.65	19206	16	16	3/8	7/8
11	56.75	51523	16	16	7/8	3/8

*For $k \leq 5$, $(4, 2, 0)$ does not exist; the entry is the node E_{k-2} .

$\mu(N_{UV})$ matches no node of the $k - 2$ model, as you found. The two ingredients of the mismatch behave differently. $\mathfrak{so}(3)_k$ lives in $\mathbb{Q}(\zeta_K)$ and the IR in $\mathbb{Q}(\zeta_{8(K-2)})$, and these fields intersect only in $\mathbb{Q}$. $\mathbb{Z}_4$ is the only piece that can be matched.

2 No-go: $\text{so}(3)_k$ cannot act on an IR that contains the $k - 2$ model

Claim. For odd $k \geq 3$, no IR phase of the form $\text{CFT}_{k-2} \oplus (\text{gapped vacua})$ carries the lines $\text{so}(3)_k$, with any number of gapped vacua.

Argument.

1. No topological interface connects $c_{k-2} > 0$ with $c = 0$. A line therefore maps the CFT vacuum to itself, $L|\Omega_{\text{CFT}}\rangle = \chi(L)|\Omega_{\text{CFT}}\rangle$. Its restriction to the CFT block is nonzero, since $L\bar{L} \ni 1$, and is a sum of lines of the $k - 2$ model. So $\chi(L) = \sum_j n_j d_j > 0$.
2. χ is then a character of the $\text{so}(3)_k$ fusion ring that is positive on every basis element. Such a character must be FPdim. Hence $d_L \in \mathbb{Q}(\zeta_K) \cap \mathbb{Q}(\zeta_{8(K-2)} \text{ or } \zeta_{24 \cdot 8(K-2)})$, whose real part is $\mathbb{Q}$ because $\gcd(K, K-2) = 1$ with K odd.
3. But $d_2 = 1 + 2\cos(2\pi/K)$ is irrational for odd $K \geq 5$. Contradiction.

The Witten index fixes the number of CFT blocks: $k + 1 = (k - 1) + 2$. A tensor-product IR $\text{CFT} \otimes \text{TQFT}_n$ would give $n(k - 1)$ Ramond ground states and is excluded.

Consequence (selection.pdf, transportability). Any flow that stays inside N_{UV} preserves F_{UV} exactly, since the lines commute with every operator of the node, the perturbed stress tensor included. Such a flow is therefore fully massive. Section 4 shows that this is the Chebyshev flow.

3 The massless flow lives in a larger node: E_k , with $\mu = 16$

The LG coupling of X^k has dimension $y_k = 2/K$, and the F-term of X^{k-2} , i.e. $(k - 2, k - 2, 2) \sim (2, -4, 0)$, has $y_{k-2} = 4/K = 2y_k$. These are resonant, so at $O(\lambda^2)$ the linear superpotential $X^{k+2} + \lambda X^k$ differs from the pure ϕ_{UV} perturbation by a coupling $g \propto \lambda^2$ to $(2, \mp 4, 0)$. That operator is charged under $\text{so}(3)_k$, since $(L, 0, 0)$ acts on $l = 2$ by a ratio $\neq d_L$. This is the CFT version of the second-order obstruction in N2Gab (§3.4, §5.2).

The Chebyshev superpotential is the unique completion that keeps $\text{so}(3)_k$ (N2Gab §4), and λ is its flat coordinate. The flow inside N_{UV} is therefore the Chebyshev one. The massless flow is instead generated by

$$N'_{UV} = \langle (0, -2, 0), (2, -4, 0) \rangle = E_k, \quad F' = \mathbb{Z}_4, \quad \mu' = 16 = \mu(E_{k-2}) = \mu(N_{IR}). \quad (4)$$

So the selection rule is satisfied as $E_k \rightarrow E_{k-2}$, and the chain $N_{UV} \subset E_k$ has relative index $\lambda = \dim \text{so}(3)_k$.

Anomaly matching. The Verlinde lines of the simple current $g = (0, K, 1)$ form $\text{Vec}_{\mathbb{Z}_4}^\omega$. For K odd, $h_g = (1 - 2K)/8 \bmod 1$, which is $7/8$ or $3/8$. So $\omega = 8h_g \bmod 4$ equals 3 for every odd k , in both the UV and the IR (table). The generator of $\mathbb{Z}_4$ is anomalous, while $\mathbb{Z}_2 = \langle (0, 0, 2) \rangle$, with $h = 1/2$, is anomaly free. The DHR categories of the two nodes, $\mathcal{Z}(\text{Vec}_{\mathbb{Z}_4}^\omega)$, therefore coincide.

On the chiral ring, g acts as $(-1)^l$, i.e. $X \rightarrow -X$. Together with the R -rotation it squares to $(-1)^s$. For odd $d = K$ this is the LG symmetry $X \rightarrow -X$, $W \rightarrow -W$, compensated by $\theta \rightarrow i\theta$. It preserves $X^d + \lambda X^{d-2}$ and exchanges the massive vacua $X = \pm i\sqrt{\lambda}$.

4 TQFT content from entropic order parameters

Work in the node E_k , with $\lambda = 4$, and use the certainty relation of entropic.pdf (3.55)/(3.60) together with its scaling analysis (§3.4, §4):

$$S_{\text{int}}(x; R) + S_{\text{tw}}(x; R) = \log 4. \quad (5)$$

- *CFT vacuum.* All sectors of E_k stay massless, because $\mu(E_k) = \mu(E_{k-2})$. The massless subalgebra of massless.pdf contains all of E_{k-2} . Both parameters approach conformal functions of x , with $U(1) = -\frac{1}{2} \log 16$ (modular.pdf (3.30)). No TQFT is required on this block.
- *Gapped vacua.* Every sector of E_k is massive. By §4 of entropic.pdf, each order parameter tends to a shape-independent constant, $S_{\text{int}} \rightarrow \log \lambda_{\text{SSB}}$ and $S_{\text{tw}} \rightarrow \log \lambda_{\text{unbr}}$, with

$$\lambda_{\text{SSB}} \lambda_{\text{unbr}} = 4. \quad (6)$$

An unbroken part must be anomaly free (it needs a fiber functor). Since $\omega(\mathbb{Z}_4) \neq 0$ and $\omega|_{\mathbb{Z}_2} = 0$, the options are $\lambda_{\text{SSB}} \in \{2, 4\}$. This gives the bound on the TQFT total dimension

$$2 \leq D_{\text{TQFT}}^2 = \lambda_{\text{SSB}} \leq 4 = \sqrt{\mu(E_k)}. \quad (7)$$

The upper bound is the certainty-relation bound $S_{\text{int}} \leq \log \lambda$. The lower bound is anomaly inflow.

- The LG count of massive vacua, 2, saturates the lower bound. **The TQFT is the $\mathbb{Z}_4 \rightarrow \mathbb{Z}_2$ SSB phase.** It has two vacua exchanged by g , unbroken $(-1)^F = (0, 0, 2)$, and topological local operators $\{1, \sigma\}$ with $\sigma \sim \text{sign}(X)$, $d_\sigma = 1$, $D^2 = 2$. In these vacua $S_{\text{int}} \rightarrow \log 2$ (constant law) and $S_{\text{tw}} \rightarrow \log 2$.

Check with the Chebyshev (pure- N_{UV}) flow. Here $\lambda = 4 \dim \mathfrak{so}(3)_k$. $\mathfrak{so}(3)_k$ has no fiber functor, since its dimensions are not integers, so it must be broken. The number of vacua n must satisfy $n \geq \deg \mathbb{Q}(d_2) = \varphi(K)/2$, because a rank- n NIM-rep needs eigenvalues of degree $\leq n$. $\mathbb{Z}_4$ is broken to $\mathbb{Z}_2$ as above. The regular module then gives $n = \frac{k+1}{2} \times 2 = k+1$ vacua and $\lambda_{\text{SSB}} \lambda_{\text{unbr}} = \dim \mathfrak{so}(3)_k \cdot 2 \cdot 2$. This reproduces exactly the $k+1$ massive vacua of the Chebyshev flow (N2Gab §4, the Witten index). The same analysis also shows why only 2 gapped vacua cannot host $\mathfrak{so}(3)_k$ for $k \geq 5$: $\varphi(K)/2 > 2$.

5 Relation to massless.pdf

In each vacuum sector, take $W_m =$ operators with $\mu_O \geq m > 0$ and A_m its commutant-type algebra.

- In the CFT vacuum of the massless flow, no sector of E_k is in W_m , so $A_m \supseteq E_{k-2}$ and no emergent symmetry is needed. The gap condition of the notes, no accumulation of masses near 0, holds: the only masses are $\sim |\lambda|^{K/4}$ from kinks to the massive block.
- The conjecture “massless and massive together $\Rightarrow$ symmetry” is realised here across vacua, not within one. The symmetry is the anomalous $\mathbb{Z}_4$. Its anomaly forbids the massive block from being a single trivially gapped vacuum and forces the two-vacuum TQFT.
- For the pure N_{UV} flow, the argument of §2 shows that no $\mathfrak{so}(3)_k$ -charged sector can remain massless. If one did, the neutral stress tensor would leave a CFT block on which $\mathfrak{so}(3)_k$ would have to act. So A_m is trivial and the flow is gapped.

Summary

1. $\langle(0, -2, 0)\rangle = \mathbb{Z}_K$ with $F = \mathfrak{so}(3)_k \boxtimes \mathbb{Z}_4$ and $\mu = 16 \dim^2 \mathfrak{so}(3)_k$. No $k-2$ node matches it, and §2 shows that no IR containing the $k-2$ model can carry $\mathfrak{so}(3)_k$. Staying in this node means the Chebyshev massive flow, with $k+1$ vacua = regular $\mathfrak{so}(3)_k$ module $\times \mathbb{Z}_4/\mathbb{Z}_2$.

2. The linear (massless) LG flow resonantly switches on $(2, \mp 4, 0)$ at $O(\lambda^2)$. The node becomes $E_k = \{(\text{even}, \text{even}, 0)\}$ with $F = \mathbb{Z}_4$ and $\mu = 16$. It flows to $E_{k-2} = \langle(4, 2, 0)\rangle$, with $\mu = 16$ and matching $\mathbb{Z}_4$ anomaly $\omega = 3$.
3. Entropic bound on the TQFT: $2 \leq D_{\text{TQFT}}^2 \leq 4$. The LG count gives $D^2 = 2$: the two massive vacua are the $\mathbb{Z}_4 \rightarrow \mathbb{Z}_2$ SSB TQFT of the anomalous $\mathbb{Z}_4$ generated by $(0, K, 1)$ ($X \rightarrow -X$ with R -rotation).

Caveats.

- Odd k only. For even k , $[k, 0, 0]$ is invertible and survives (N2Gab App. D), so F' is larger.
- The identification “flow inside $N_{UV} = \text{Chebyshev}$ ” uses the uniqueness of N2Gab §4 within their ansatz, plus flat-coordinate = CFT-coupling.
- The field-theoretic statement in §2 assumes that the IR lines commuting with the image node have dimensions in the cyclotomic field of the $k - 2$ model, which holds for lines of that rational model.
- The anomaly class is read off from the simple-current spin, $\omega = 8h_g \bmod 4$.